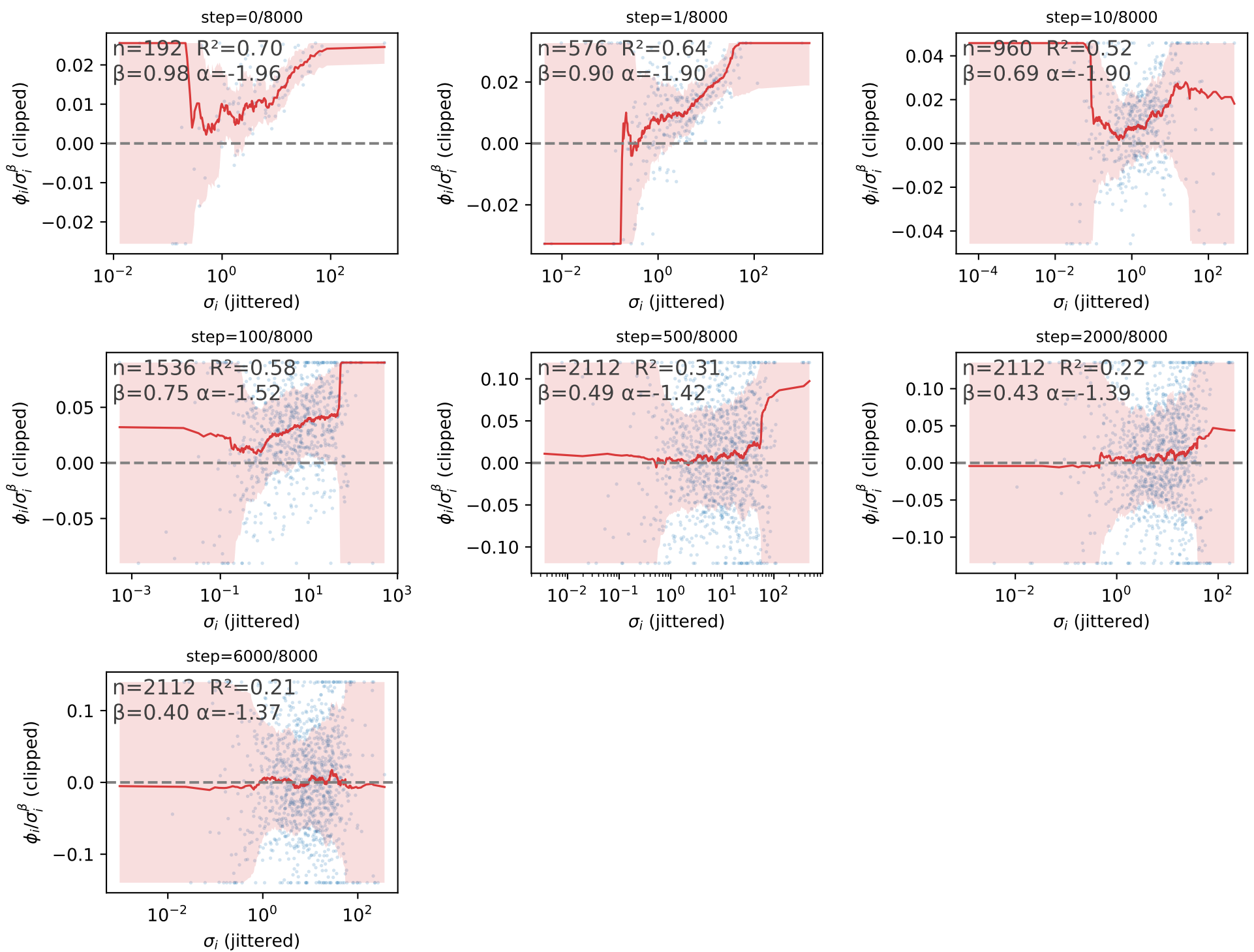
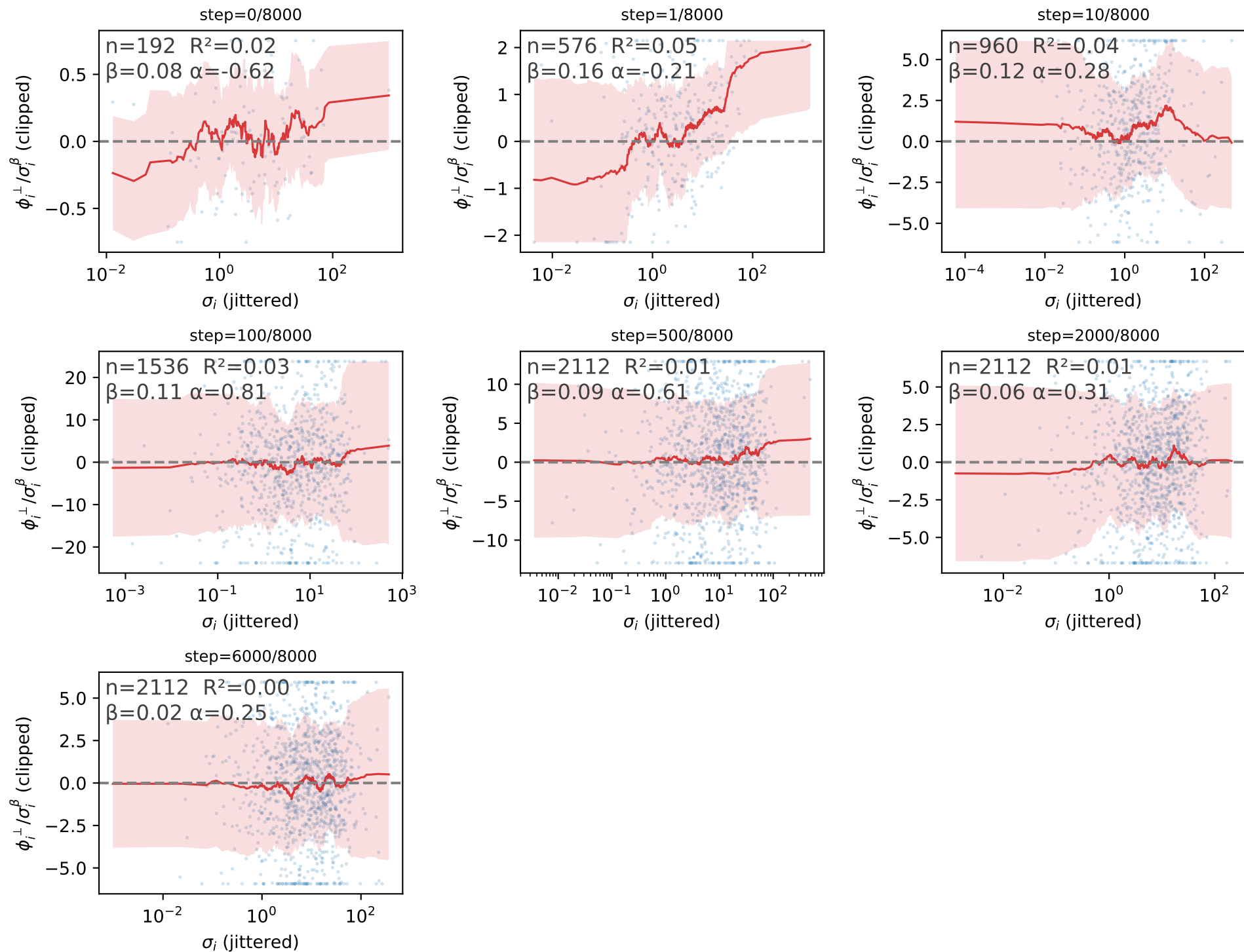


Attention value projection: ϕ_i/σ_i^β vs σ_i (β fit per group from $\log|\phi_i| \sim \beta \log \sigma_i$, one batch-mean point per mode per step in the group)



n: number of points.
 R^2 , β , α : least-squares fit $\log_{10}(y) = \beta \cdot \log_{10}(x) + \alpha$ (β the fitted power-law exponent, α the intercept).

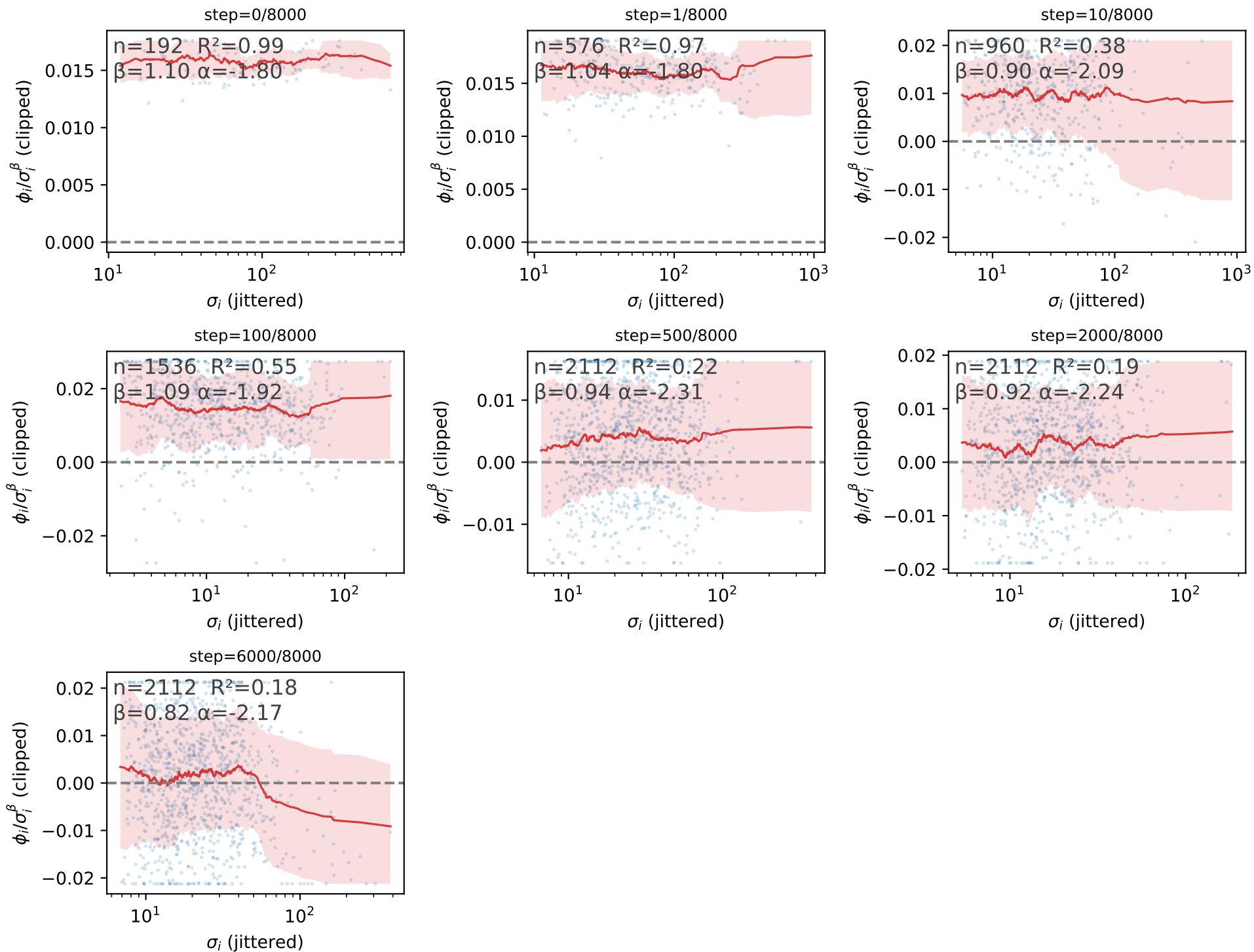
Attention value projection: $\phi_i^\perp / \sigma_i^\beta$ vs σ_i (β fit per group from $\log|\phi_i^\perp| \sim \beta \log \sigma_i$, one batch-mean point per mode per step in the group)



n: number of points.

R^2 , β , α : least-squares fit $\log_{10}(y) = \beta \cdot \log_{10}(x) + \alpha$ (β the fitted power-law exponent, α the intercept).

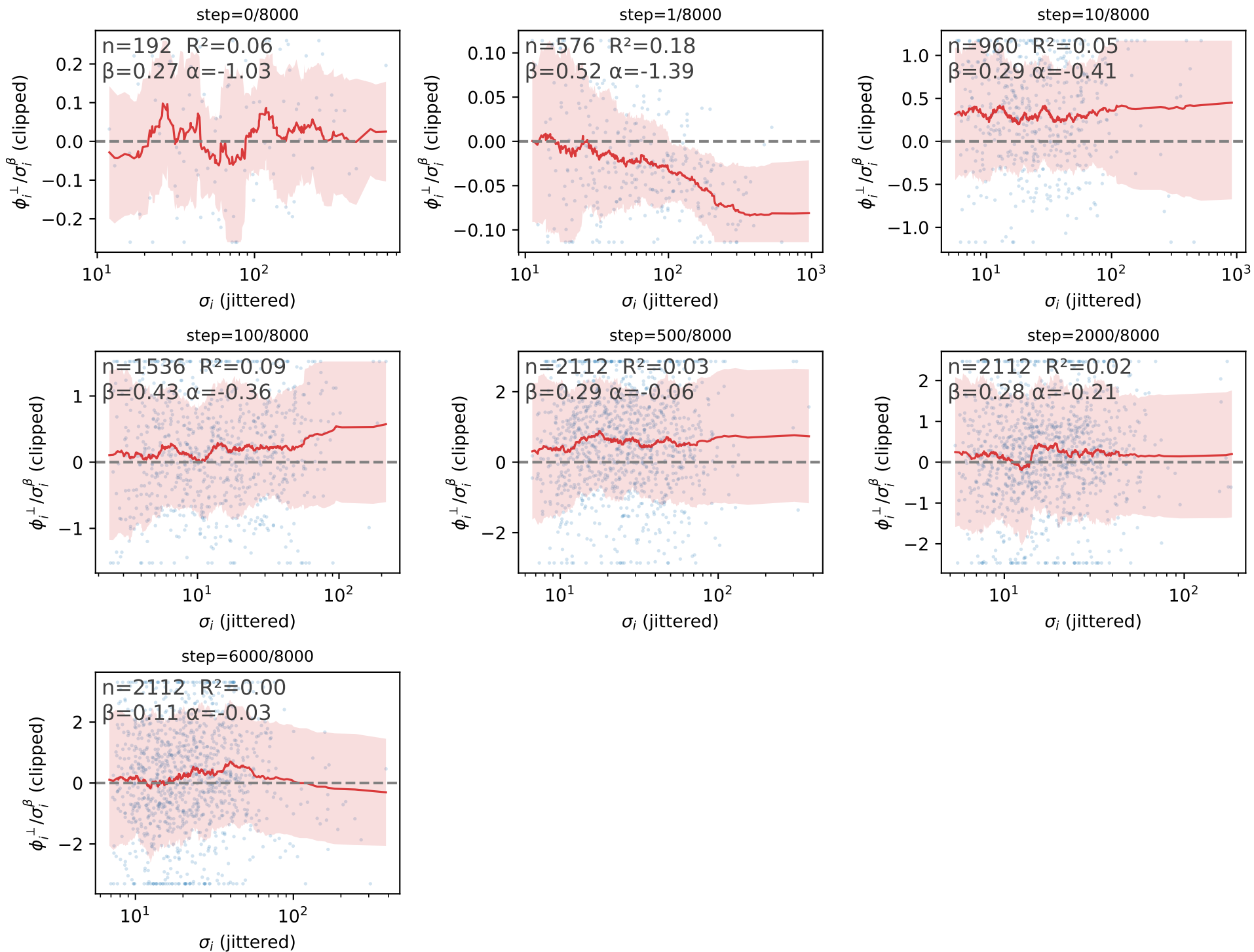
MLP up-projection: ϕ_i/σ_i^β vs σ_i (β fit per group from $\log|\phi_i| \sim \beta \log \sigma_i$, one batch-mean point per mode per step in the group)



n : number of points.

R^2 , β , α : least-squares fit $\log_{10}(y) = \beta \cdot \log_{10}(x) + \alpha$ (β the fitted power-law exponent, α the intercept).

MLP up-projection: $\phi_i^\perp/\sigma_i^\beta$ vs σ_i (β fit per group from $\log|\phi_i^\perp| \sim \beta \log \sigma_i$, one batch-mean point per mode per step in the group)



n: number of points.

R^2 , β , α : least-squares fit $\log_{10}(y) = \beta \cdot \log_{10}(x) + \alpha$ (β the fitted power-law exponent, α the intercept).